
Feedback or Annealing?

A Controlled Study of Adaptive State Aggregation

Sunny Zhang
University of Toronto
ssunny.zhang@mail.utoronto.ca

Abstract

Adaptive state aggregation replaces full-state Bellman updates with cheaper updates on groups of states with similar estimated values. A feedback rule narrows the groups as the Bellman residual shrinks, but any gain may come from the narrowing schedule rather than the residual signal. I separate these effects on a 9,261-state inventory control problem. Because each full-state sweep reduces the residual range to at most γ times its previous value, the feedback rule has a roughly geometric schedule that can be fixed in advance. I compare the two methods under three update mixtures, 20 paired seeds, and five compute budgets. Among comparisons whose 95% bootstrap interval excludes zero, feedback’s largest advantage is 0.2% of the baseline error and its largest disadvantage is 10.8%. After 10,000 iterations, every interval contains zero and lies within the ± 0.02 range specified before the experiment as negligible. By contrast, gradual narrowing changes error by about 23% at one matched budget, helping early but hurting later. Full-state value iteration also reaches the reference tolerance with fewer counted Bellman backups. Evaluations should therefore compare feedback with fixed-width and matched preplanned baselines, measure policy quality, and use several compute budgets.

1 Introduction

State aggregation reduces the cost of dynamic programming by solving a smaller approximation to a Markov decision process (MDP) [1–3]. In Chen et al. [4], states with similar current values share a group; full-state updates alternate with cheaper updates that sample one state per group. A fixed group width ε trades cost against accuracy: wide groups are cheaper, while narrow groups reduce approximation error.

The Bellman residual measures the change from another full update. It suggests using wide groups when estimates are changing rapidly and narrower groups as they stabilize. However, comparison with a fixed width cannot distinguish the value of the residual signal from the value of narrowing groups over time.

I compare feedback with a geometric schedule fixed in advance and matched, for each update mix, on its starting width, minimum width, and time to that minimum. Thus, only feedback differs. I retain the inconclusive result for the comparison specified before the experiment, one mix at 20 ms, and add fixed-computation comparisons at three mixes. The Bellman contraction property justifies the geometric comparison (Proposition 1). Across five budgets, feedback never improves on it, although gradual narrowing changes error by about 23% relative to a fixed width and changes from helpful to harmful as the budget grows.

2 Related work

Classical aggregation assigns one value to each state group; its errors depend on within-group value variation [1, 3, 6]. Adaptive methods group by residual size [2], adjust partial membership [7], divide regions using value, policy, influence, or variance [8], or divide them as the value function changes [9]. I use Chen et al. [4], which groups states by current value and has asymptotic error bound $2\varepsilon/(1-\gamma)$.

Residuals give error bounds [10], have been minimized directly [11], and guide state selection in prioritized sweeping [12, 13]. Because a small residual need not imply a good policy [14], I also measure policy loss. Here the residual range changes only group width, not membership rules, state priorities, or the fitting objective; no new representation or convergence theorem is proposed.

3 Method

For a finite discounted cost MDP, the Bellman update T maps the current value function V to

$$(TV)(s) = \min_a \left[c(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \right]. \quad (1)$$

The solver groups values $V(s)$ into nonempty intervals of width ε . A full-state iteration evaluates (1) for every state. A grouped iteration samples one state from each group, evaluates one Bellman update for that state, and uses the result to update the shared group value with step size $\alpha_t = 1/\sqrt{t}$. A schedule (ℓ_g, ℓ_a) repeats ℓ_g full-state iterations followed by ℓ_a grouped iterations. The groups are rebuilt before each grouped phase. Their values are copied back to individual states before the next full-state phase.

After each full-state sweep, I record the residual range $r_t = \max_s (TV_t - V_t)(s) - \min_s (TV_t - V_t)(s)$. The tested feedback rule is

$$\varepsilon_t^{\text{RESIDUAL}} = \max\{\varepsilon_{\min}, cr_t\}, \quad c = 0.084, \quad \varepsilon_{\min} = 0.05. \quad (2)$$

For the (2, 5) mix specified before the experiment, the constant was fixed to align the first residual-controlled width with the coarsest fixed baseline: there the first grouped phase follows $\ell_g = 2$ common opening sweeps, after which $r = 5.93429$ deterministically, so $0.084r = 0.49848 \approx 0.5$. The opening sweeps are ordinary value iteration, so this width is available before the feedback method is run. It changes with ℓ_g , giving $\varepsilon_0^{\text{RESIDUAL}} = 0.400, 0.499$, and 0.538 at (5, 2), (2, 5), and (1, 20). The main control never reads r_t :

$$\varepsilon_j^{\text{GEOMETRIC}} = \max \left\{ \varepsilon_{\min}, \varepsilon_0^{\text{RESIDUAL}} \left(\frac{\varepsilon_{\min}}{\varepsilon_0^{\text{RESIDUAL}}} \right)^{j/(C-1)} \right\}, \quad (3)$$

over grouped phases j , with $(\varepsilon_0^{\text{RESIDUAL}}, C)$ set per mix to (0.400, 7), (0.499, 13), and (0.538, 14): the starting width, the minimum, and the cycle at which RESIDUAL reaches the minimum. GEOMETRIC therefore matches the start and end of RESIDUAL's schedule at every mix. The only difference is that RESIDUAL uses the residual to choose each width, whereas GEOMETRIC follows a schedule fixed in advance. A method with fixed $\varepsilon = 0.05$ tests whether gradual refinement is useful at all. Results for slower geometric schedules and coarser fixed widths are included in the supplementary material.

$\varepsilon_0^{\text{RESIDUAL}}$ is predictable in advance but C is not, so GEOMETRIC obtains it from one RESIDUAL run. At the two added mixes, GEOMETRIC is therefore a matched comparison not specified before the experiment. This adjustment changes only the comparison method and makes its schedule more similar to RESIDUAL, which makes any advantage for feedback harder to find. The control specified before the experiment used (0.4985, 14) for every mix. At (2, 5), that schedule agrees with the matched control to within one cycle. Section 5 shows the effect of using the unmatched schedule at the other mixes.

Why use a geometric comparison? The start and end of GEOMETRIC are matched to RESIDUAL, and a standard result determines the general rate between them.

Proposition 1 (contraction of the residual range, 5, §6.6). *During the full-state sweeps of any schedule, $r_{t+1} \leq \gamma r_t$. Hence $\varepsilon^{\text{RESIDUAL}} \leq \max\{\varepsilon_{\min}, \varepsilon_0^{\text{RESIDUAL}} \gamma^{n_g}\}$, where n_g counts full-state sweeps since*

the first grouped phase, and the width reaches its minimum within $\lceil \log(\varepsilon_{\min}/\varepsilon_0^{\text{RESIDUAL}}) / \log \gamma \rceil$ of them.

This contraction result is standard [5, §6.6]; I do not claim it as new. Appendix A includes the proof to make the constants explicit. Its consequence for ε is what matters here. Because the feedback rule reads a residual that shrinks geometrically during full-state sweeps, a schedule fixed in advance can reproduce its general rate. The proposition does not cover grouped-update phases, which also change V . It therefore provides an upper bound only while the residual is still setting the width. The bound holds at every recorded step in that period for all three mixes. Per cycle, $\varepsilon^{\text{RESIDUAL}}$ changes by factors of 0.698, 0.829, and 0.829, compared with upper bounds γ^{ℓ_g} of 0.774, 0.903, and 0.950. Grouped updates account for the gap between these numbers. For that reason, the exact cycle C must be measured rather than calculated from γ^{ℓ_g} alone.

Inventory MDP, protocol, and metrics. A state is the inventory vector $q \in \{-10, \dots, 10\}^3$, giving $|\mathcal{S}| = 21^3 = 9,261$. Five choices of quote aggressiveness trade fill probability against spread, and the one-period cost combines spread revenue with correlated inventory risk (Appendix C). I use $\gamma = .95$ and rescale costs so $\|V^*\|_\infty = 100$. This is a small market-making-style control model, not a calibrated market simulator. Each method runs for 10,000 iterations at each mix. The MDP itself is deterministic, so seeds 0–19 vary only which state is sampled within each group. The same seeds are used for all methods. The main accuracy measure is $\text{err}_\infty = \|V - V^*\|_\infty$, the largest absolute error across states. I also evaluate the policy chosen from V using $\|V^{\pi_V} - V^*\|_\infty$. Plots against computation use counted Bellman backups $B = n_g|\mathcal{S}| + \sum_t K_t$, where K_t is the number of occupied groups. This count favors aggregation because it omits the residual scan and the $(n_r + n_l)|\mathcal{S}|$ state operations used to rebuild and expand groups. Paired differences are summarized by their median and a 95% percentile bootstrap interval based on 10,000 resamples. Before the experiment, I defined differences in $[-0.02, 0.02]$ on the 100-normalized value scale as too small to matter in practice.

4 Results

Once the residual and geometric rules reach $\varepsilon_{\min}$ their error curves are visually indistinguishable at every mix (Figure 1, Appendix B). Table 1 shows the small numerical differences. At all three mixes, the 95% interval for residual minus matched geometric both contains zero and stays within ± 0.02 . The median differences are 1.7×10^{-7} , 5.1×10^{-7} , and -0.00201 . Thus, when the comparison has the same starting width, minimum width, and time to that minimum, using the residual does not improve the result after 10,000 iterations.

Value error and policy quality need not agree. At (2, 5), the residual and matched geometric methods have the same value error to seven decimal places, yet their policy losses differ by 19%. At (1, 20), the residual method is worse than fixed-0.05 on both measures: its value error is 0.01250 higher (95% CI [0.00745, 0.01561]), and its policy loss is 0.03529 higher. Neither measure supports using feedback in this experiment. Within each mix, the median computation at the end differs by at most 0.13%, although rebuilding and expanding the groups adds 39–96% more operations for the residual method. Full-state VI reaches its 10^{-10} tolerance in 454 sweeps, or 4.20M counted backups. Aggregation is therefore not competitive on this example; the experiment studies one approximation method rather than proposing a better solver.

Separating the schedule from the feedback. Results after 10,000 iterations hide gradual narrowing because each scheduled method has used $\varepsilon_{\min}$ for most of the run; at (5, 2), residual and fixed end only 3.8×10^{-8} apart. Table 2 instead compares four earlier budgets while widths still change. It reports relative differences because a ± 0.02 threshold is uninformative when errors are near 10.

Table 1: Results after 10,000 iterations over 20 paired seeds. R/G/F denote residual, matched geometric, and fixed $\varepsilon = .05$; policy loss and B(M), millions of counted backups, use the same order. The interval summarizes within-seed differences.

Mix	$\text{err}_\infty(\text{R})$	$\text{err}_\infty(\text{G})$	$\text{err}_\infty(\text{F})$	R–G (95% CI)	Policy loss R/G/F	B(M) R/G/F
(5, 2)	.05001	.05001	.05001	$1.7\text{e}-7$ [$-2.8\text{e}-7, 9.0\text{e}-7$]	.00210/.00210/.00210	67.685/67.684/67.686
(2, 5)	.13882	.13882	.13792	$5.1\text{e}-7$ [$-9.6\text{e}-5, 2.0\text{e}-5$]	.01518/.01280/.01280	30.214/30.206/30.195
(1, 20)	.20343	.20115	.19093	-.00201 [$-.00320, .00541$]	.05178/.05178/.01649	9.185/9.186/9.196

Table 2: Median paired value-error difference at fixed budgets, as a percentage of the comparison error; negative favors RESIDUAL. RESIDUAL–FIXED measures gradual narrowing and RESIDUAL–GEOMETRIC the added effect of feedback. * marks a 95% bootstrap interval excluding zero; details are in Appendix D.

Mix	Contrast	10^5	3×10^5	10^6	3×10^6	final
(5, 2)	RESIDUAL–FIXED	+0.0	+0.2*	−0.1	−0.0	+0.0
(5, 2)	RESIDUAL–GEOMETRIC	+0.0	+0.0	−0.2*	+0.0	+0.0
(2, 5)	RESIDUAL–FIXED	+0.1	−23.5*	−14.1*	+0.4*	+0.6*
(2, 5)	RESIDUAL–GEOMETRIC	−0.1	−0.1	+0.2*	−0.0	+0.0
(1, 20)	RESIDUAL–FIXED	−18.8*	−5.1*	−5.4*	+10.2*	+6.5*
(1, 20)	RESIDUAL–GEOMETRIC	+4.5*	+10.8*	+3.4*	−1.4	−1.0

Among intervals excluding zero, feedback’s largest advantage is 0.2% of the comparison error and its largest disadvantage is 10.8%. Gradual refinement changes error by about 23%: at (1, 20) it is 18.8% better than a fixed width after 10^5 backups but 10.2% worse after 3×10^6 . Thus, the schedule changes both the result and its direction, while feedback adds no consistent benefit, in line with Proposition 1. Large RESIDUAL–FIXED differences occur during rapid error reduction; their direction and scale are stable, but exact percentages depend on which update the budget reaches.

The timing-based primary measure was inconclusive. At 20 ms, the original (2, 5) primary comparison gave a residual-minus-geometric median of -0.0417 with 95% CI $[-0.0459, 0.0011]$, which is inconclusive under the specified equivalence rule. Repeated runs selected different iterations because error was falling rapidly and machine load varied. I retain this outcome; the backup-based comparisons were added later and were not specified before the experiment.

5 What the negative result shows

The schedules must be matched on the right scale. A preplanned comparison must use the unit that triggers width changes, here grouped phases rather than total iterations. Copied unchanged to (1, 20), the slow geometric schedule never reaches its minimum and ends ten times worse; correcting only its cycle count removes the difference (Appendix E).

A mismatched schedule creates an apparent effect. Fixed at (0.4985, 14) for every mix, the geometric method specified before the experiment differs from residual at (5, 2) by .00632, 95% CI [.006319, .006327], with all 20 seeds favoring geometric. The mismatch causes this precise difference: geometric starts 25% wider and takes twice as long to reach the minimum. Matching both features reduces the difference to 1.7×10^{-7} , with an interval containing zero. Low simulation variance can therefore make a mismatched comparison look like a feedback effect.

6 Limitations and conclusion

One deterministic instance, residual statistic, minimum width, and feedback constant cannot support claims about other MDPs. The mixes, budget ladder, and backup measurements were not specified before the experiment, and the backup count omits group maintenance. Feedback may still help when regional residuals differ, the best minimum width is unknown, or computation ends before refinement does.

Within the scope of this experiment, residual feedback never improves on a preplanned schedule with the same start, minimum, and duration. Gradually narrowing the groups does matter, changing error by about 23% at one matched budget, but its effect changes sign. Attributing an improvement to feedback requires a matched preplanned schedule and several compute budgets.

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A Proof of Proposition 1

The following standard result says that the Bellman operator shrinks differences when they are measured by their range [5, §6.6]. The proof is included to make the constants in the bound explicit.

Fix V, U and a state s , and let a^* attain the minimum in $(TU)(s)$. Evaluating the bracket of (1) at a^* rather than at the minimizer for V gives

$$\begin{aligned} (TV)(s) &\leq c(s, a^*) + \gamma \sum_{s'} P(s' | s, a^*) V(s') = (TU)(s) + \gamma \sum_{s'} P(s' | s, a^*) (V - U)(s') \\ &\leq (TU)(s) + \gamma \max_{s'} (V - U)(s'), \end{aligned}$$

and the symmetric argument, taking a^* to attain the minimum in $(TV)(s)$, gives $(TV)(s) \geq (TU)(s) + \gamma \min_{s'} (V - U)(s')$. Both bounds hold at every s , so $(TV - TU)(s)$ lies in $[\gamma \min(V - U), \gamma \max(V - U)]$. Defining $\text{span}(x) = \max(x) - \min(x)$, it follows that

$$\text{span}(TV - TU) \leq \gamma \text{span}(V - U).$$

During a full-state phase $V_{t+1} = TV_t$, so $r_{t+1} = \text{span}(TV_{t+1} - V_{t+1}) = \text{span}(TV_{t+1} - TV_t) \leq \gamma \text{span}(V_{t+1} - V_t) = \gamma \text{span}(TV_t - V_t) = \gamma r_t$. Iterating over the n_g full-state sweeps taken since the first grouped phase and substituting into (2) gives the stated bound, and the minimum width is reached once $\varepsilon_0^{\text{RESIDUAL}} \gamma^{n_g} \leq \varepsilon_{\min}$.

The proposition does not cover a grouped phase. Such a phase copies updates from representative states to their groups instead of applying T to every state, so it can increase the residual range. The bound therefore applies only while the rule is still setting the width. It holds empirically at every recorded step in that period for all three mixes.

B Error over computation

Rules reaching $\varepsilon_{\min}$ have similar error floors — inventory $N = 3$, $Q = 10$, $\gamma = 0.95$

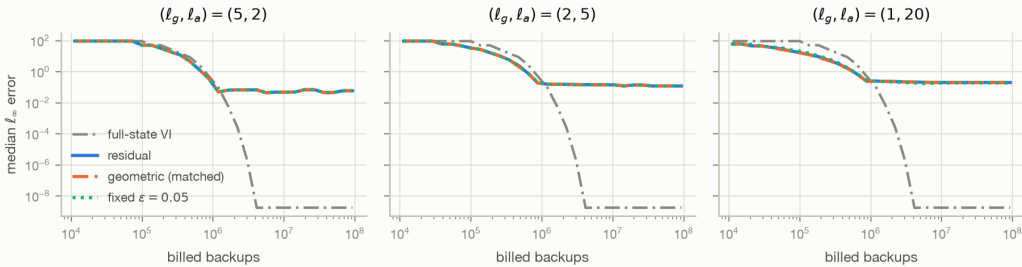


Figure 1: Median value error against counted Bellman backups. Final estimates are displayed beyond each method’s 10,000-iteration limit. Residual, fast geometric, and fixed-0.05 nearly coincide within each mix. The deterministic VI curve is repeated for reference; its convergence check stops after 454 sweeps (4.20M backups) though the plotted diagnostic continues to the common stopping point.

The flat right-hand portions repeat the final values; they do not represent continued computation. Each method stops after 67.7M, 30.2M, or 9.2M counted backups at (5, 2), (2, 5), or (1, 20), respectively. The plots carry the last value forward so that the three panels can share an axis. Budgets beyond a method’s stopping point therefore cannot be compared fairly, which is why the list in Appendix D stops at 3×10^6 .

C Inventory instance

The five joint quote-aggressiveness actions have fill probabilities (.1, .2, .4, .6, .8) and spreads (1, .8, .5, .3, .15), so aggressive quotes fill often at a worse price. The one-period cost is

$$c(q, a) = 0.02 q^\top \Sigma q - \delta_a f_a L(q),$$

where δ_a and f_a are the spread and fill probability of action a , Σ has unit diagonal and correlation 0.5 so inventory risk is correlated across the three assets, and $L(q)$ removes fills that would leave the box $\{-10, \dots, 10\}^3$. Costs are rescaled after solving so that $\|V^*\|_\infty = 100$, which puts ε , the equivalence margin, and every reported error on one interpretable scale.

D How the budget comparisons are calculated

Each cell of Table 2 is the median over 20 seeds of the within-seed difference divided by the comparison method’s error for that seed at the same budget. Calculating the ratio before taking the median preserves the pairing. Errors are read as a step function of counted backups: each budget uses the most recent completed update. The list stops at 3×10^6 because the cheapest mix, (1, 20), performs only 9.2M backups in total. Later budgets would compare a repeated final value with continued computation. The supplementary material reports absolute differences and 10,000-resample percentile-bootstrap intervals for both the absolute and relative medians.

E Adjusting the slow schedule across update mixes

The slow geometric method specified before the experiment used 1,429 grouped-update cycles, exactly the number induced by 10,000 iterations at cycle length seven. Transplanted unchanged to (1, 20), whose cycle length is 21, it receives only 477 grouped phases and is still narrowing at the stopping point. It ends at $\varepsilon = .232$ and error 1.59, an order of magnitude worse than every other method. Recalculating only its cycle count to 477, changing nothing else, ends it at $\varepsilon = .05024$ and error .209, back in the neighborhood of residual and fast geometric. The poor result came from an unfinished schedule, not from the lack of feedback. For this reason, mixes other than the (2, 5) mix specified before the experiment use the schedule with the corrected cycle count.